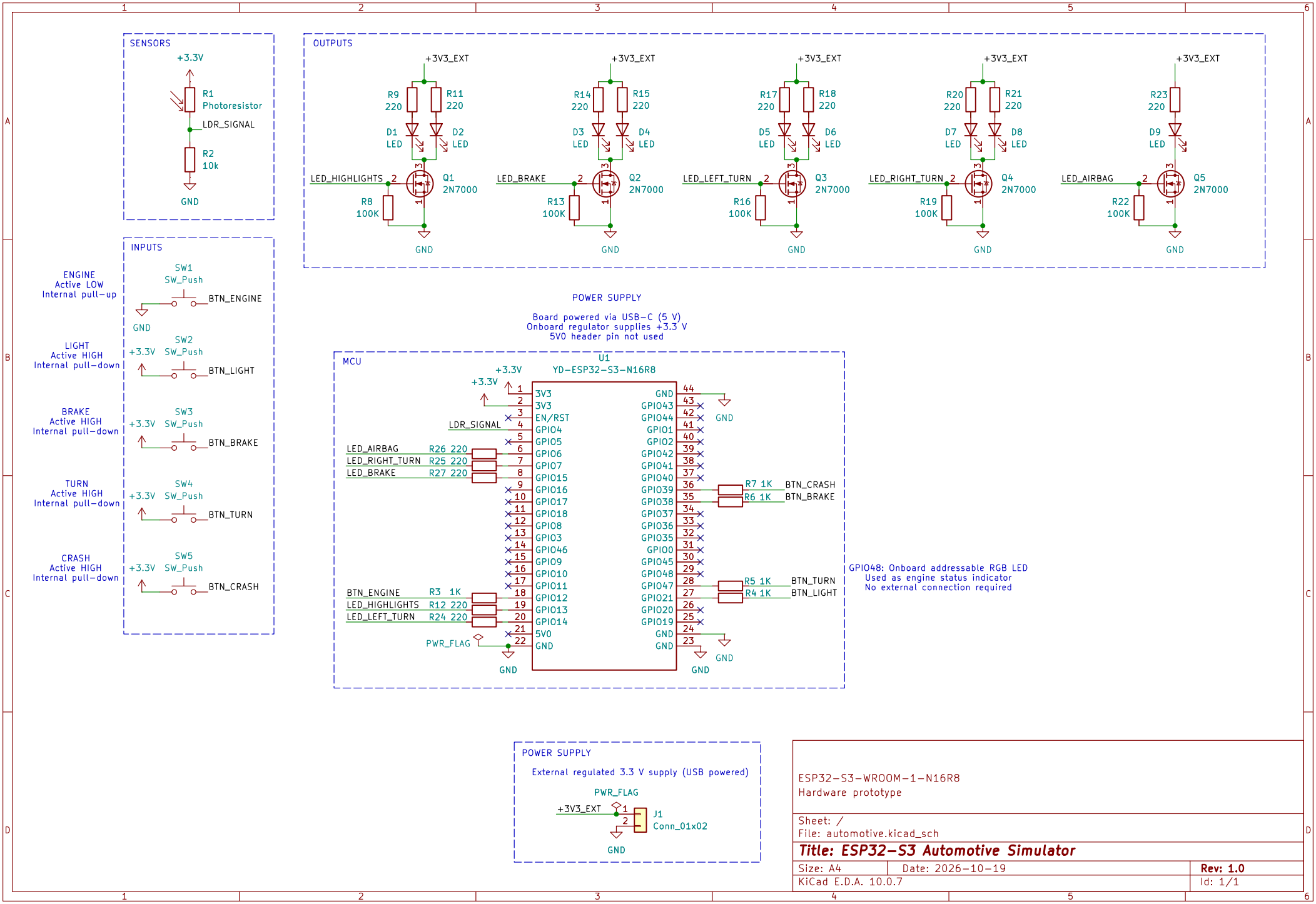

SENSORS
+3.3V
R1
Photoresistor
LDR_SIGNAL
R2
10k
GND
INPUTS
ENGINE
Active LOW
Internal pull-up
SW1
SW_Push
BTN_ENGINE
GND
LIGHT
Active HIGH
Internal pull-down
SW2
+3.3V
SW_Push
BTN_LIGHT
BRAKE
Active HIGH
Internal pull-down
SW3
+3.3V
SW_Push
BTN_BRAKE
TURN
Active HIGH
Internal pull-down
SW4
+3.3V
SW_Push
BTN_TURN
CRASH
Active HIGH
Internal pull-down
SW5
+3.3V
SW_Push
BTN_CRASH
OUTPUTS
+3V3_EXT
R9 220
R11 220
D1 LED
D2 LED
LED_HIGHLIGHTS
Q1 2N7000
R8 100K
GND
R14 220
R15 220
D3 LED
D4 LED
LED_BRAKE
Q2 2N7000
R13 100K
R17 220
R18 220
D5 LED
D6 LED
LED_LEFT_TURN
Q3 2N7000
R16 100K
R20 220
R21 220
D7 LED
D8 LED
LED_RIGHT_TURN
Q4 2N7000
R19 100K
R23 220
D9 LED
LED_AIRBAG
Q5 2N7000
R22 100K
POWER SUPPLY
Board powered via USB-C (5 V)
Onboard regulator supplies +3.3 V
5V0 header pin not used
MCU
U1
YD-ESP32-S3-N16R8
+3.3V
3V3
EN/RST
GPIO4
GPIO5
GPIO6
GPIO7
GPIO15
GPIO16
GPIO17
GPIO18
GPIO8
GPIO3
GPIO46
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
5V0
GND
GPIO43
GPIO44
GPIO1
GPIO2
GPIO42
GPIO41
GPIO40
GPIO39
GPIO38
GPIO37
GPIO36
GPIO35
GPIO0
GPIO45
GPIO48
GPIO47
GPIO21
GPIO20
GPIO19
LDR_SIGNAL
LED_AIRBAG R26 220
LED_RIGHT_TURN R25 220
LED_BRAKE R27 220
BTN_ENGINE R3 1K
LED_HIGHLIGHTS R12 220
LED_LEFT_TURN R24 220
PWR_FLAG
R7 1K BTN_CRASH
R6 1K BTN_BRAKE
R5 1K BTN_TURN
R4 1K BTN_LIGHT
GPIO48: Onboard addressable RGB LED
Used as engine status indicator
No external connection required
POWER SUPPLY
External regulated 3.3 V supply (USB powered)
PWR_FLAG
+3V3_EXT
J1
Conn_01x02
GND
ESP32-S3-WROOM-1-N16R8
Hardware prototype
Sheet: /
File: automotive.kicad_sch
Title: ESP32-S3 Automotive Simulator
Size: A4
Date: 2026-10-19
Rev: 1.0
KiCad E.D.A. 10.0.7
Id: 1/1